

KUVEMPU UNIVERSITY

Jnanasahyadri, Shankaraghatta, Shivamogga - 577451.



“Internship on Full Stack Web Development”

An internship report in partial fulfillment of the requirements for the award of degree of

BACHELOR OF COMPUTER APPLICATIONS

Submitted by

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U06GE23S0008



Government First Grade Evening College

Bapuji Nagara, Shivamogga – 577201

2025-2026

**Government First Grade Evening College
Bapuji Nagara, Shivamogga – 577201**

Department of Computer Science

CERTIFICATE



*This is to certify that the internship work entitled **Internship on Full Stack Web Development submitted** in partial fulfillment of the requirement for the award of the degree **BACHELOR OF COMPUTER APPLICATIONS** from Kuvempu University, Shankaragatta, is a result of the bonafide work carried out*

By

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during the academic year 2025-2026.

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DECLARATION

I, **Mohammed Afaf Hassan**, student of 6th semester BCA, **Government First Grade Evening College, Bapuji Nagar, Shivamogga**, hereby declare that the internship entitled “**Internship On Full Stack Web Development** ” has been independently carried out by me under the supervision of external guide **Vinay M V, VIN TUTORIALS, Shivamogga** and internal guide **Chethan N J, Lecturer**, department of Computer Science, Government First Grade Evening College, Bapuji Nagar, Shivamogga and submitted in partial fulfillment of the requirements for the award of degree in **BACHELOR OF COMPUTER APPLICATIONS** by the Kuvempu University during the academic year 2025-2026.

Place: Shivamogga
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Acknowledgement

Every successful completion of any undertaking would be complete only after I remember and thank the almighty, the parents, the teachers and all the personalities, who directly or indirectly helped and guided during the execution of that work. The success of this work is equally attributed to all well-wishers who have encouraged and guided throughout the execution. I would like to express my sincere thanks to Principal, Prof. SHASHIDHAR. S, GFGECE, Shivamogga for encouraging me and giving all the facilities.

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GUIDE CERTIFICATE

*This is to certify that the Internship work entitled “**Internship on Full Stack Web Development**” carried out by **Mohammed Afaf Hassan**, 6th Semester **BCA** under my guidance towards the partial fulfillment of the requirement for the award of the degree **BACHELOR OF COMPUTER APPLICATIONS** from Kuvempu University, Shankaraghatta, during the academic year 2025-2026.*

Place: Shivamogga

Date:

Chethan N J

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INTRODUCTION

1.1 INTRODUCTION OF INTERNSHIP:

The internship in Full Stack Web Development provided practical knowledge and hands-on experience in designing, developing, and managing web applications. Full Stack Web Development involves both front-end and back-end technologies required to build complete web-based systems. During this internship, I learned how to create responsive user interfaces using HTML, CSS, JavaScript, and Bootstrap, as well as how to manage server-side operations using PHP and database management using MySQL.

As part of the internship, I worked on the project titled “Online Examination System”, which is a web-based application developed to conduct exams digitally in an efficient and secure manner. The system helps students attend exams online, view results, and allows administrators to manage questions, users, and exam records easily.

This internship helped me understand real-time project development, problem-solving, database connectivity, user authentication, and system testing. It improved my technical skills, teamwork, and confidence in developing professional web applications. The experience gained during this internship plays an important role in building my career in the field of web development and software engineering.

1.2 PURPOSE OF INTERNSHIP:

The main purpose of the internship in Full Stack Web Development was to gain practical knowledge and real-time experience in developing web applications using both front-end and back-end technologies. It helped in understanding how a complete web-based system is designed, developed, tested, and maintained in a professional environment.

Through this internship, I aimed to improve my technical skills in HTML, CSS, JavaScript, Bootstrap, PHP, and MySQL. The internship also provided an opportunity to apply theoretical knowledge learned in academics to a real project, which is the “Online Examination

System”.

The purpose was also to understand project planning, database management, user authentication, error handling, and system implementation. It helped in improving problem-solving abilities, communication skills, teamwork, and confidence in handling software development projects.

Overall, the internship was conducted to prepare for future career opportunities in web development and to gain industry-level exposure in building practical software solutions.

INTERNSHIP ON PHP

➤ Full Stack Web Development – Online Examination System

Full Stack Web Development is the process of developing both the front-end and back-end of a web application. It includes designing the user interface, handling server-side logic, managing databases, and deploying the complete application. In this internship, the project developed was an Online Examination System using modern web technologies.

The Online Examination System helps students and administrators conduct examinations digitally. It simplifies the process of creating exams, managing questions, attending tests, calculating results, and storing student performance records without manual paperwork.

➤ During this internship, the following technologies were used:

- HTML – for page structure
- CSS – for styling
- JavaScript – for validation
- PHP – backend logic
- MySQL – database
- XAMPP – local server

➤ HTML

HTML stands for Hypertext Markup Language. It is the standard markup language used to create the structure of web pages. In the Online Examination System, HTML is used to design different pages such as home page, login page, registration page, exam page, result page, admin dashboard, and student dashboard.

HTML provides elements like forms, buttons, input fields, tables, images, and headings which help in creating the user interface. In this project, HTML forms are used to collect student details such as name, registration number, course details, login credentials, and exam information.

HTML helps in organizing content in a structured format and acts as the backbone of the web application. It connects the front-end with the back-end and makes the layout of the Online Examination System possible.

➤ CSS

CSS stands for Cascading Style Sheets. It is used to design and style web pages. CSS improves the appearance of the Online Examination System by adding colors, fonts, spacing, alignment, and layout. It helps in making the website attractive and user-friendly.

In this project, CSS is used to design the homepage, exam forms, buttons, navigation bar, and result pages. It also helps in maintaining consistency across all pages such as login page, exam page, and dashboard.

CSS improves readability and gives a professional look to the Online Examination System, making it easier for students and administrators to use the system.

➤ JavaScript

JavaScript is a client-side scripting language used to add interactivity to web pages. It is mainly used for validation and dynamic behaviour. In the Online Examination System, JavaScript is used to validate user input before sending data to the server.

For example, JavaScript checks whether required fields are filled, whether registration number is valid, whether exam answers are submitted properly, and whether login details are correct.

It prevents invalid data submission and improves system reliability. JavaScript also helps in displaying alert messages, timer functions during exams, confirmation messages, and result notifications, which improves user experience and reduces server load.

➤ PHP

PHP stands for Hypertext Preprocessor. It is a server-side scripting language used to develop dynamic web applications. PHP plays a major role in the Online Examination System. It handles backend logic such as student registration, login authentication, exam management, result calculation, and report generation.

In this project, PHP connects to MySQL database and performs operations like inserting student details, updating exam records, managing questions, and retrieving result history.

PHP also manages sessions to keep students logged in during examination sessions. When a student enters login details, PHP verifies the information from the database. If correct, the student is redirected to the dashboard.

PHP makes the Online Examination System functional, secure, and dynamic

➤ MySQL

MySQL is a relational database management system used to store data. In the Online Examination System, MySQL is used to store student details, exam questions, answer records, marks, result details, and exam history.

Database tables are created to store student information and exam details. When a student attends an exam, the database is updated. When administrators view results, data is retrieved from the database.

MySQL ensures data consistency, secure storage, and efficient management of records. It also supports queries like SELECT, INSERT, UPDATE, and DELETE which are used in this project.

➤ XAMPP

XAMPP is a local server environment used to run PHP and MySQL projects. It includes Apache server, MySQL database, PHP interpreter, and phpMyAdmin. XAMPP allows developers to run web applications on a local system without internet connection.

In this project, XAMPP is used to start Apache and MySQL services. Project files are stored inside the htdocs folder. Database is created using phpMyAdmin. When the browser runs localhost, the Online Examination System loads and executes PHP code.

➤ XAMPP makes development easy and provides a testing environment for the project.

Overall, HTML, CSS, JavaScript, PHP, MySQL, and XAMPP together are used to develop the Online Examination System. These technologies help in designing the interface, validating input, processing examination logic, storing data, and running the application successfully

PROJECT DESIGN

The Online Examination System is a web-based application developed to automate and simplify the process of conducting examinations online. The main objective of this project is to provide students and administrators with an easy platform for exam management, question handling, result generation, and performance tracking. This system is developed using PHP and MySQL technologies and runs on a local server using XAMPP.

The online examination system helps in managing exams efficiently in real-time. It provides secure login authentication and allows students to attend exams from anywhere. The system stores student information, exam details, questions, and results in a database. Each examination is evaluated automatically and maintains result history.

This project provides practical knowledge about web development, database connectivity, session management, and result processing. The system is designed to be simple, user-friendly, and efficient.

3.1 Problem Definition

Traditional examination systems require manual paper handling, checking, and result preparation, which consumes more time and effort. Students must be physically present for exams, and teachers need extra time for evaluation. To reduce these problems, the Online Examination System is used to automate examination services.

The Online Examination System is developed to conduct exams in a web-based environment. The system allows users to login using username and password. After login, students can attend exams, submit answers, check results, and view performance history.

The system ensures secure access and maintains student and examination details. This project helps in understanding how online examination systems work and how results are processed using databases.

➤ The system allows users to:

- Student Registration
- Login using username
- Enter Password
- Attend Online Exam
- Submit Answers
- View Result
- Check Score
- View Exam History
- Change Password
- Logout Securely

This project simulates a real online examination process using PHP and MySQL.

3.2 Requirement Analysis

➤ Hardware Requirements

Hardware: A standard PC/Laptop (i3 processor or higher), 4GB RAM.

The hardware requirements for the Online Examination System are minimal. The system

can run on a basic computer or laptop. The following hardware components are required:

- Computer or Laptop
- Minimum 4GB RAM
- Keyboard and Mouse
- Monitor
- Internet Browser

These hardware components are sufficient to develop and run the Online Examination System. The project does not require any special hardware.

➤ **Software Requirements**

Software:

Server: XAMPP or WAMP (Apache Server)

Language: PHP 7.x or 8.x

Database: MySQL

Frontend: HTML5, CSS3, Bootstrap

The software requirements for the Online Examination System include programming languages and development tools. The following software are used:

- HTML for page structure
- CSS for styling
- JavaScript for validation
- PHP for backend logic
- MySQL for database
- XAMPP server for local hosting
- Web browser like Chrome
- VS Code or Notepad++ editor

These software tools help in developing the complete Online Examination System. XAMPP server is used to run PHP and MySQL locally. PHP connects with database and performs operations.

3.3 Design:

➤ Design

The design phase describes the structure and working of the Online Examination System. The system is divided into frontend design and backend design. The design also includes ER diagram and DFD diagram to represent database structure and data flow.

➤ Frontend (Client-Side)

The frontend is the part of the system that users interact with directly. It provides the interface for students and administrators to access the online examination system.

➤ Technologies Used

- **HTML:** Structures the web pages and content
- **CSS:** Styles the layout and improves visual appearance
- **JavaScript:** Adds interactivity and dynamic behavior

➤ Frontend Features

- User-friendly login and registration pages
- Dashboard for students and admin
- Display of exam questions and options
- Timer-based exam interface
- Submission of answers
- Result display and score visualization
- Responsive design for mobile and desktop devices

➤ Backend (Server-Side)

The backend is responsible for processing data, handling business logic, and managing communication between the frontend and database.

➤ Technologies Used

- PHP: Handles server-side logic and processing
- MySQL: Stores user data, questions, and exam results

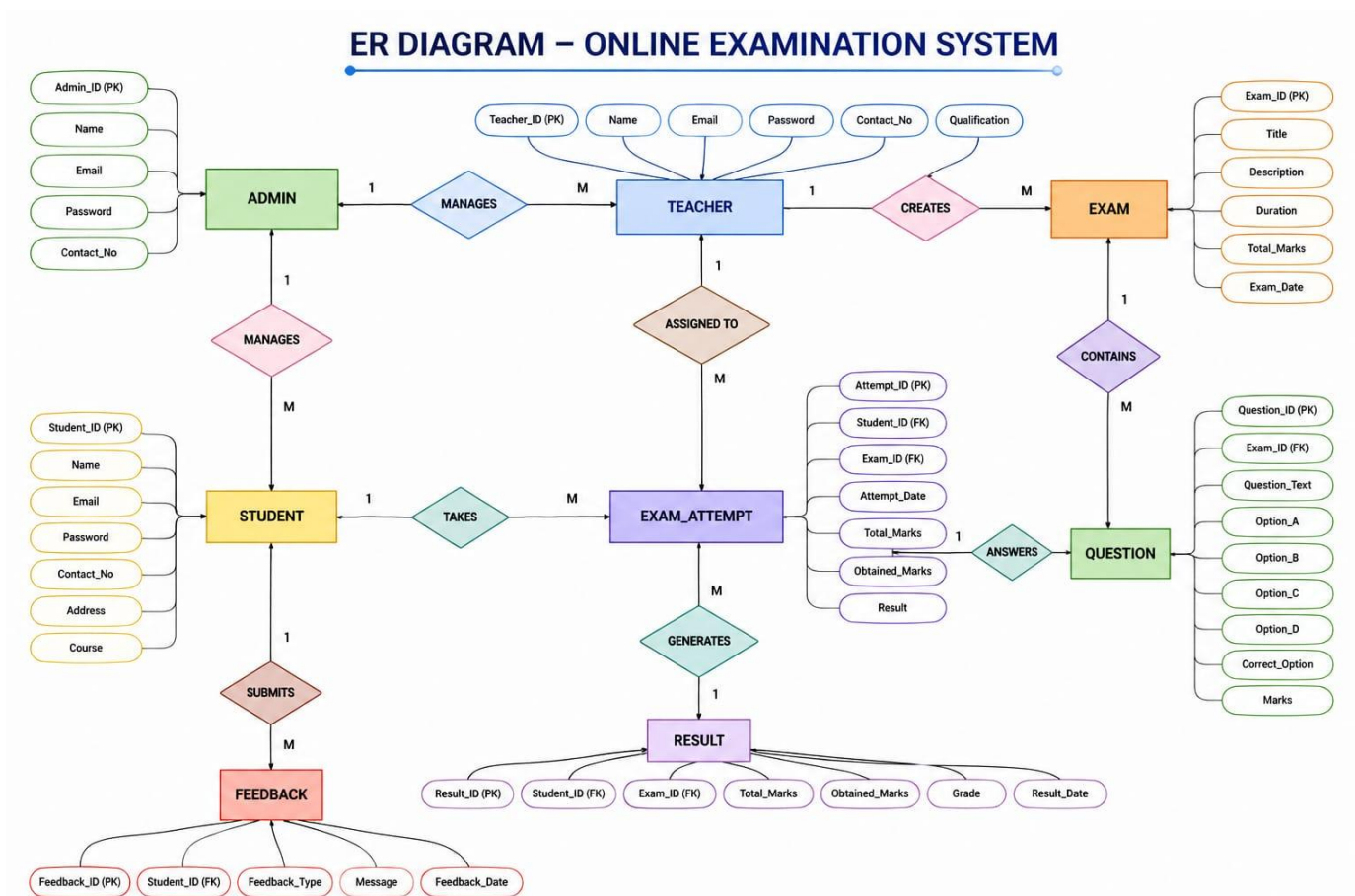
➤ Backend Features

- User authentication and authorization
- Question and exam management (add, update, delete)
- Secure storage of student data and answers
- Evaluation of answers and result generation
- API handling for frontend-backend communication
- Session management and security handling

3.7 ER DIAGRAM:

The ER Diagram includes the main entities such as Admin, Teacher, Student/User, Exam, Questions, Results, and Feedback.

Admin manages teachers and exams, teachers create questions, students attend exams, results are generated after submission, and feedback is collected from users.



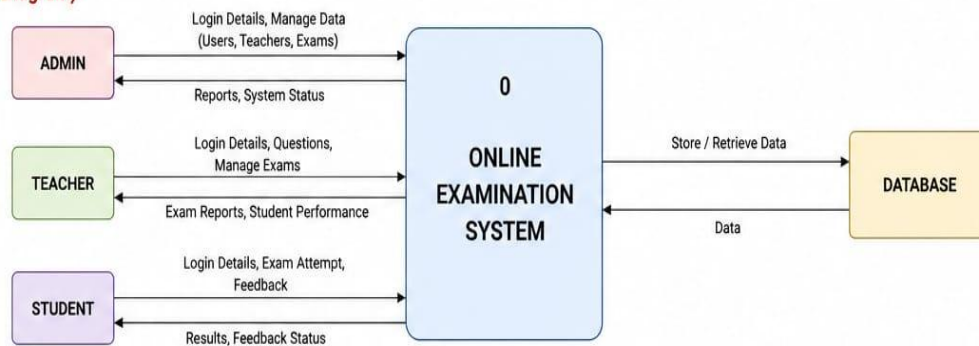
3.8 DFD (Data Flow Diagram):

The Data Flow Diagram represents the movement of data between the user, system, and database.

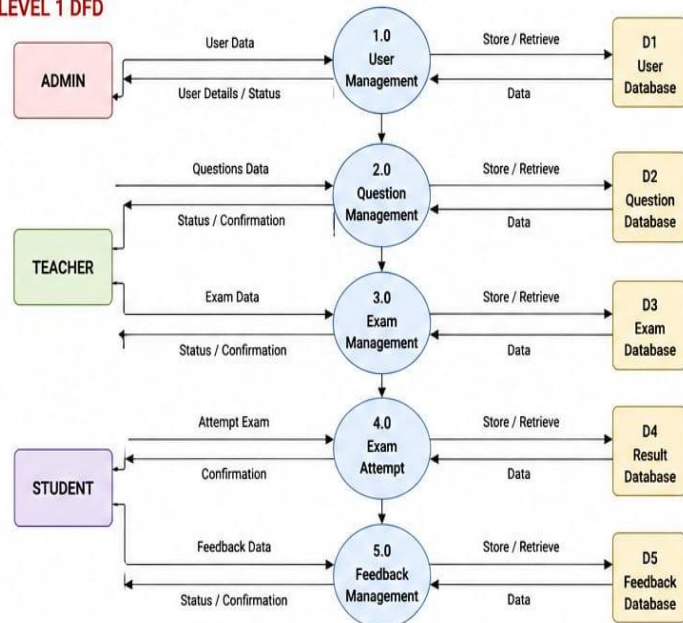
Users register and log in to the system, participate in exams, and receive results. The database stores user details, exam records, results, and feedback.

DATA FLOW DIAGRAM (DFD) – ONLINE EXAMINATION SYSTEM

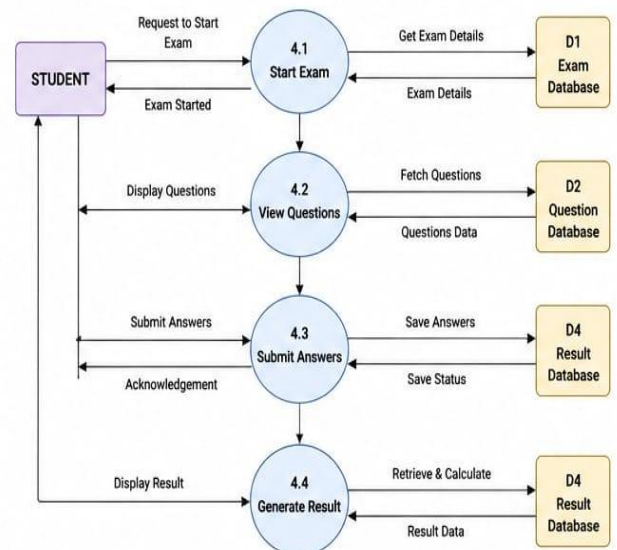
LEVEL 0 DFD (Context Diagram)



LEVEL 1 DFD



LEVEL 2 DFD – (Process 4.0 – Exam Attempt)



Legend: □ Process □ External Entity □ Data Store → Data Flow

3.9 Module – Implementation:

➤ Admin Module

The admin module is responsible for overseeing and controlling the overall functioning of the Online Examination System. It provides administrative access to manage students, exams, questions, and monitor system activities.

➤ Key Functions:

- * Monitoring all registered students and their activities
- * Adding, updating, and deleting exams
- * Managing question bank (add/edit/delete questions)
- * Viewing student results and performance
- * Removing invalid or inappropriate data
- * Managing user accounts and system settings

➤ Working:

The admin interacts with the backend through secured login credentials. The backend processes admin requests and performs operations directly on the database such as storing exam data, updating questions, and retrieving student performance records.

➤ Limitation:

The current implementation may use basic authentication. It may lack advanced security features like role-based access control and encryption.

➤ User Module (Student Module)

The User module is the core functional part of the system, enabling students to interact with the platform and attend online examinations.

➤ Key Functions:

- * Registering as a new student
- * Login using username and password

- * Viewing available exams
- * Attending online exams
- * Submitting answers
- * Viewing results and scores
- * Checking exam history
- * Updating profile details

➤ **Working:**

Students interact with the frontend interface, which sends requests to the backend. The backend processes these requests, evaluates answers, updates the database, and returns results to the frontend for display.

➤ **Features:**

- * User-friendly dashboard
- * Real-time exam interface
- * Automatic result generation
- * Personalized exam history
- * Secure login and session management

➤ **Database Module**

The Database module manages the storage, retrieval, and organization of all application data using MySQL.

➤ **Data Tables:**

- **Users Table:**

Stores student details such as username, email, password, and profile information.

- **Exams Table:**

Stores exam details such as exam name, duration, and total marks.

- **Questions Table:**

Stores questions, options, and correct answers.

- **Results Table:**

Stores student exam results, scores, and performance data.

➤ Key Functions:

- * Storing structured data in tables
- * Retrieving data based on queries (e.g., exam list, results)
- * Updating records after exam submission
- * Maintaining relationships between users, exams, and results

➤ Working:

The backend communicates with the MySQL database using PHP. SQL queries are executed to insert, update, and fetch data based on user or admin actions.

➤ I Learned How To:

- * Develop a complete web-based application
- * Manage database using MySQL
- * Implement login and session management
- * Create and manage online exams
- * Automatically evaluate answers and generate results
- * Connect frontend with backend using PHP

3.10 Testing:

Testing is a crucial phase in software development used to verify that the system functions correctly, meets requirements, and is free from critical errors. In this project, testing was performed to ensure that both frontend and backend components of the Online Examination System operate reliably and that exam data is processed accurately.

➤ Types of Testing Performed

1. Functional Testing

Functional testing was conducted to verify that all features of the system work as intended.

***Tested functionalities include:**

- * User registration and login
- * Creating and managing exams (admin)
- * Attempting online exams (students)
- * Viewing exam results and scores

- * Timer-based exam submission

Each function was tested with valid and invalid inputs to ensure correct behavior.

2. Integration Testing

Integration testing was carried out to ensure proper communication between:

- * Frontend (User Interface)
- * Backend (Server-side logic)
- * Database (Student, exam, and result data)

API endpoints were tested to confirm that data flows correctly between components and responses are accurately reflected in the user interface.

3. Database Testing

Database operations were tested to verify:

- * Correct storage of user and exam data
- * Accurate retrieval of questions and results
- * Proper updates during exam submission
- * Secure handling of student records

Consistency of data was checked after each operation.

4. User Interface Testing

The frontend interface was tested to ensure:

- * Proper rendering of exam pages and questions
- * Responsive design across devices (mobile, tablet, desktop)
- * Smooth navigation between sections (login, exam, result)
- * Clear display of timers and instructions

User interactions were verified for usability and ease of access.

5. Error Handling Testing

The system was tested for handling incorrect inputs and unexpected scenarios such as:

- * Invalid login credentials
- * Network interruptions during exams
- * Missing or incomplete answers
- * Server-side errors

➤ Testing Table

TEST CASE	EXPECTED RESULT	STATUS
User registration	New user account created successfully	Passed
User login	Valid user redirected to dashboard	Passed
Admin login	Admin redirected to admin panel	Passed
Create exam	Exam created and stored in database	Passed
Add question	Question added successfully to exam	Passed
Start exam	Exam page with timer displayed	Passed
Submit exam	Answer submitted and stored successfully	Passed
View exam	Correct score display after submission	Passed
Invalid login	Access denied with error displayed	Passed
Empty form submission	Validation message displayed	Passed

3.10 Output Screenshots:

- Home Page
- Admin Login Page
- Sign In
- Teacher Login Page
- About Page

These pages demonstrate the complete working of the Online Examination System.

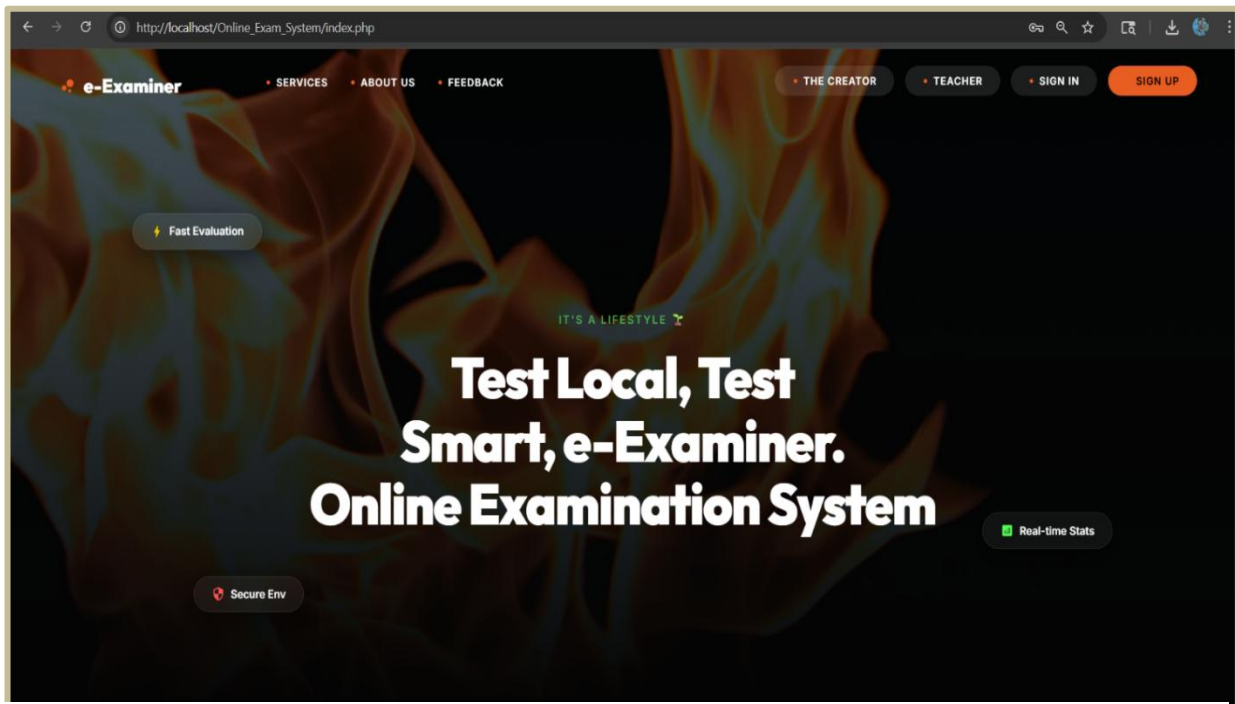


Figure 1: Home Page

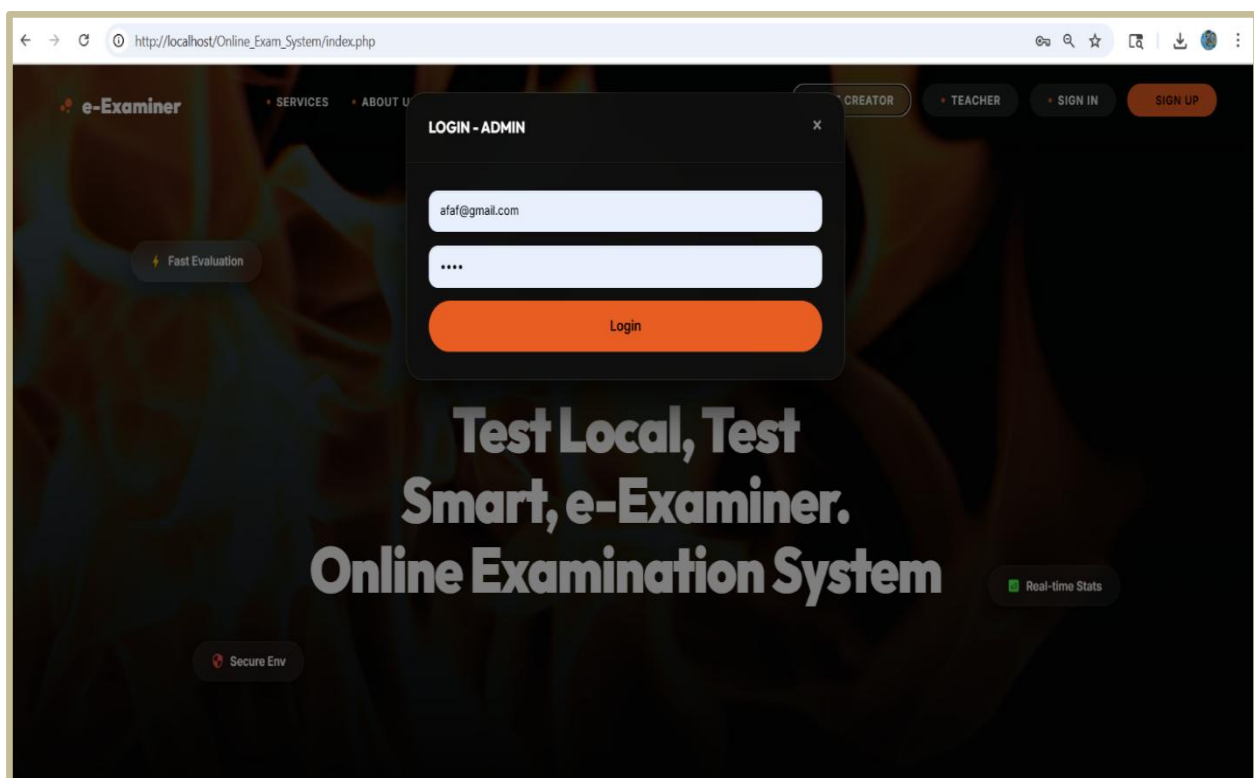


Figure 2: Admin Login

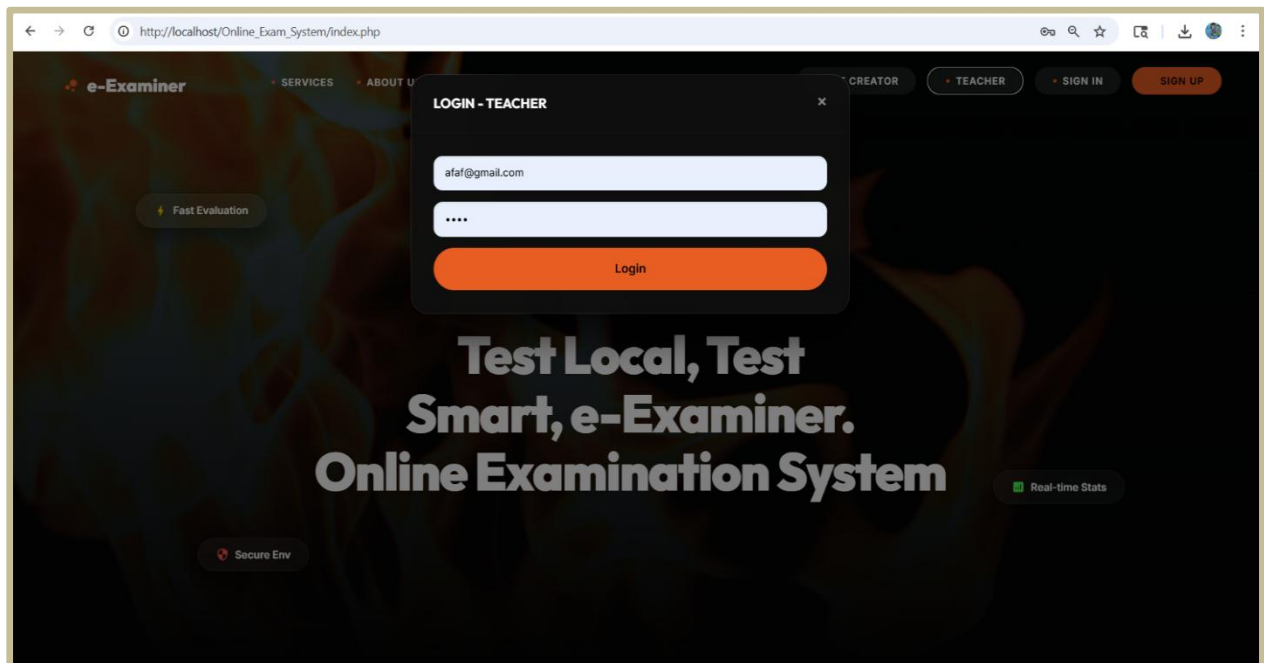


Figure 1:Teacher Login

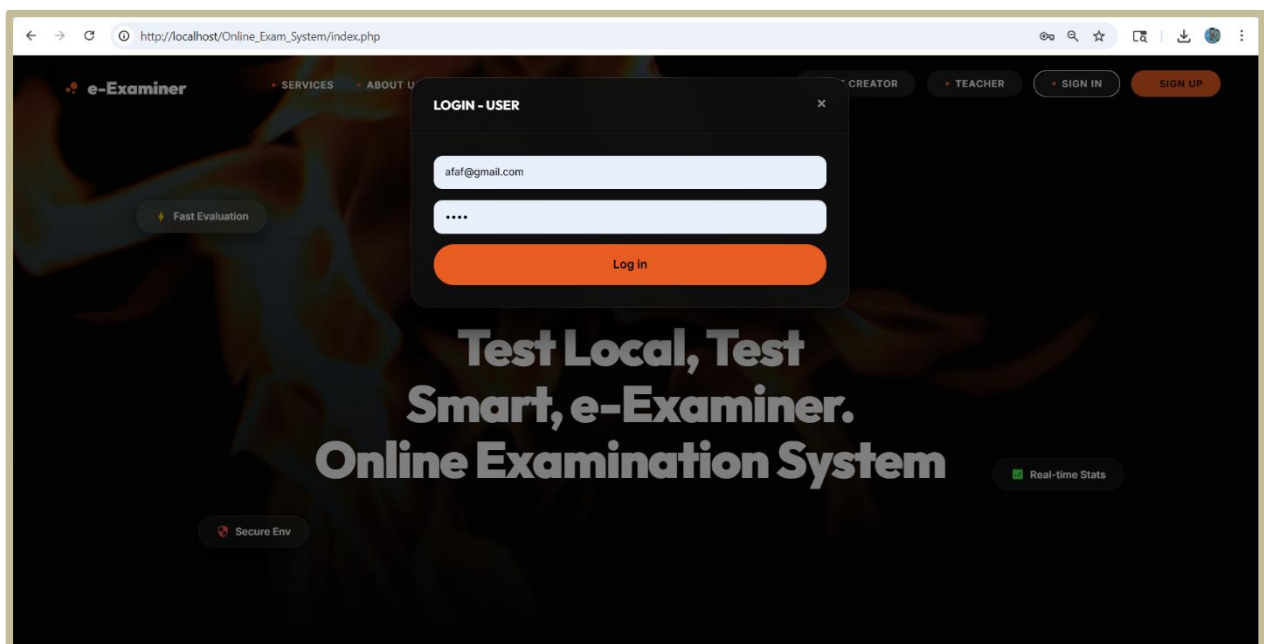


Figure 2:User Login

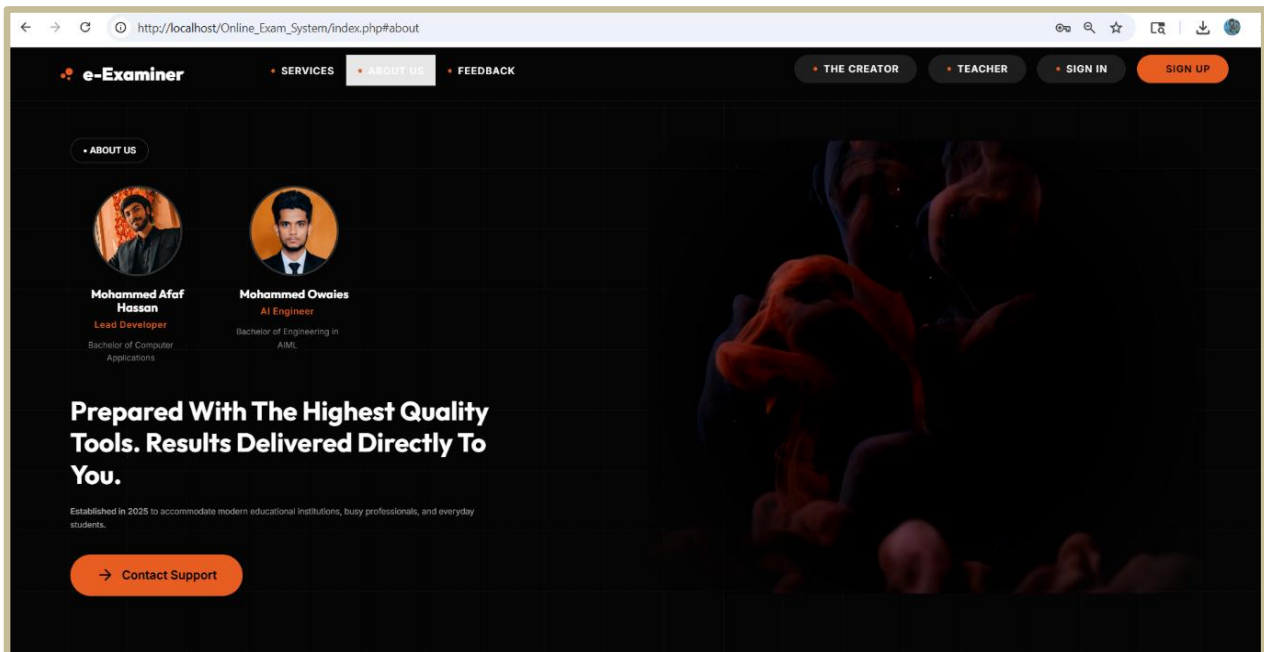


Figure 3: About Us

➤ Future Enhancement

- The system can be further improved by adding more advanced and user-friendly features to enhance the overall experience. Continuous updates and improvements will help the system adapt to modern educational requirements.
- Future versions of the Online Examination System can include enhanced security features such as two-factor authentication, data encryption, and secure browser integration to prevent malpractice and ensure data protection.
- The system can be upgraded to include advanced question types such as subjective, descriptive, and coding-based questions along with automatic evaluation support.
- Artificial Intelligence (AI) can be integrated to enable automated proctoring, face detection, and suspicious activity monitoring during online exams.

- A mobile application version of the system can be developed to provide better accessibility and convenience for students and administrators.
- Additional features such as real-time notifications, email alerts, and exam reminders can be implemented to improve communication between students and administrators.
- The system can also include analytics and reporting tools to help institutions analyze student performance, exam trends, and generate detailed reports.
- Integration with Learning Management Systems (LMS) can be enhanced to allow seamless course management, exam scheduling, and result publishing.

➤ Reference

1. PHP Official Documentation
2. MySQL Official Documentation
3. W3Schools Tutorials
4. XAMPP Documentation
5. HTML, CSS, JavaScript Learning Resources
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